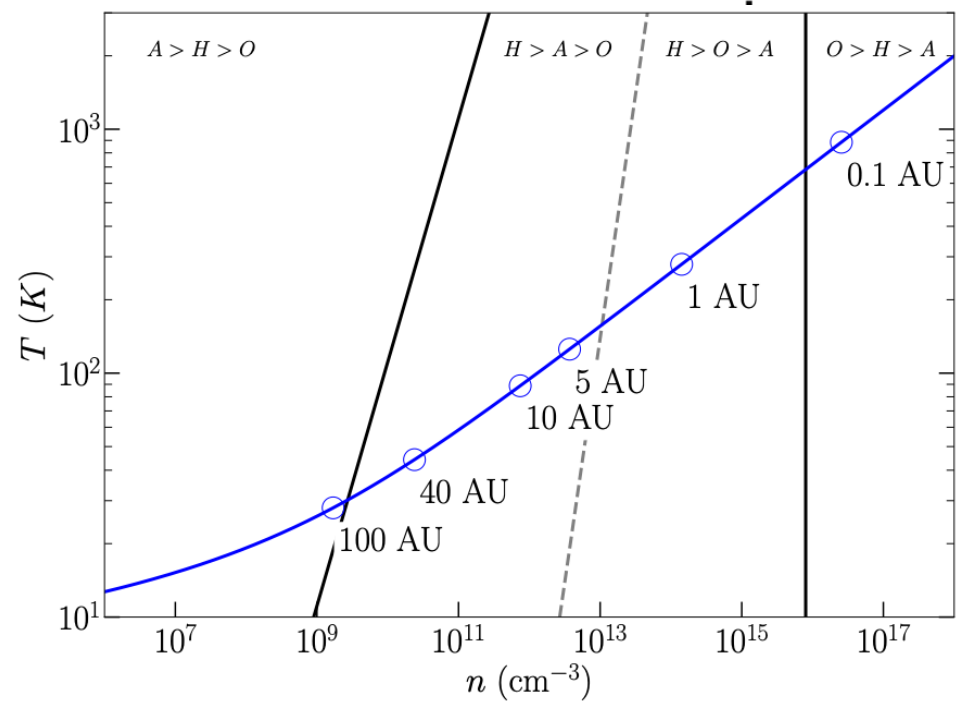
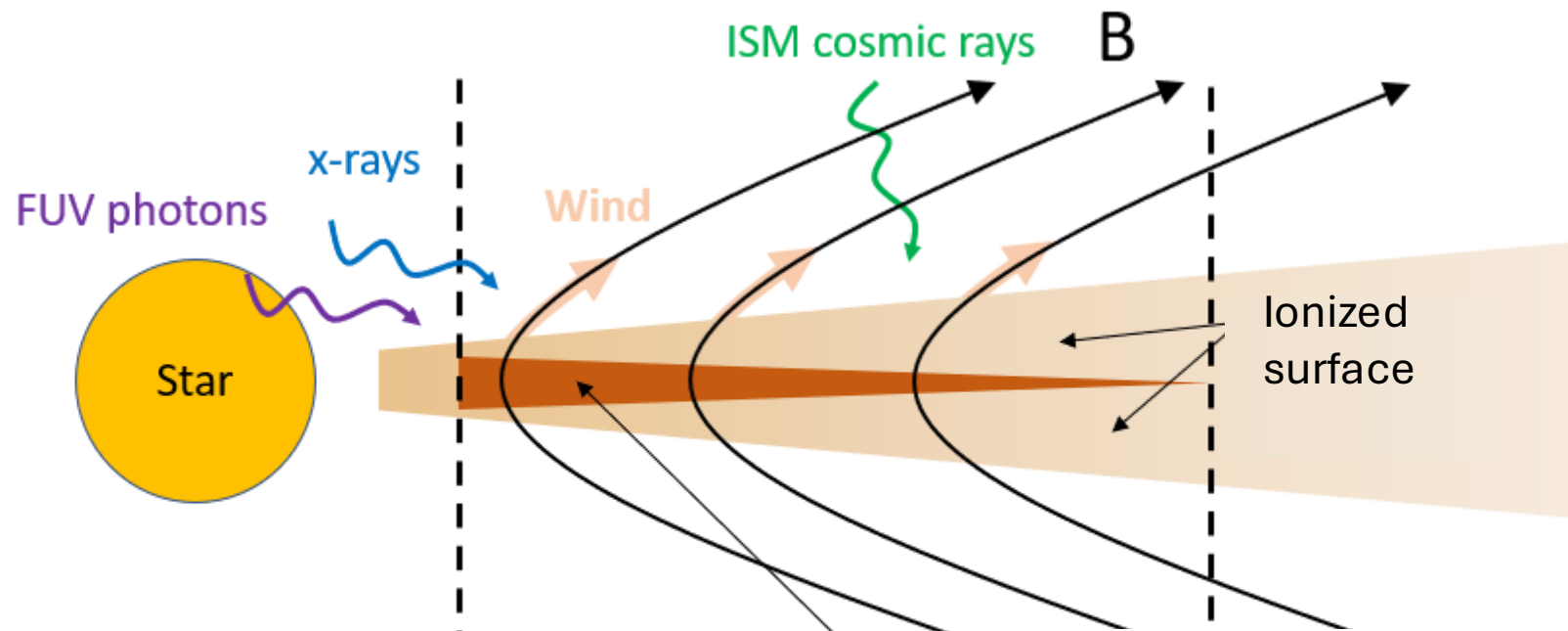
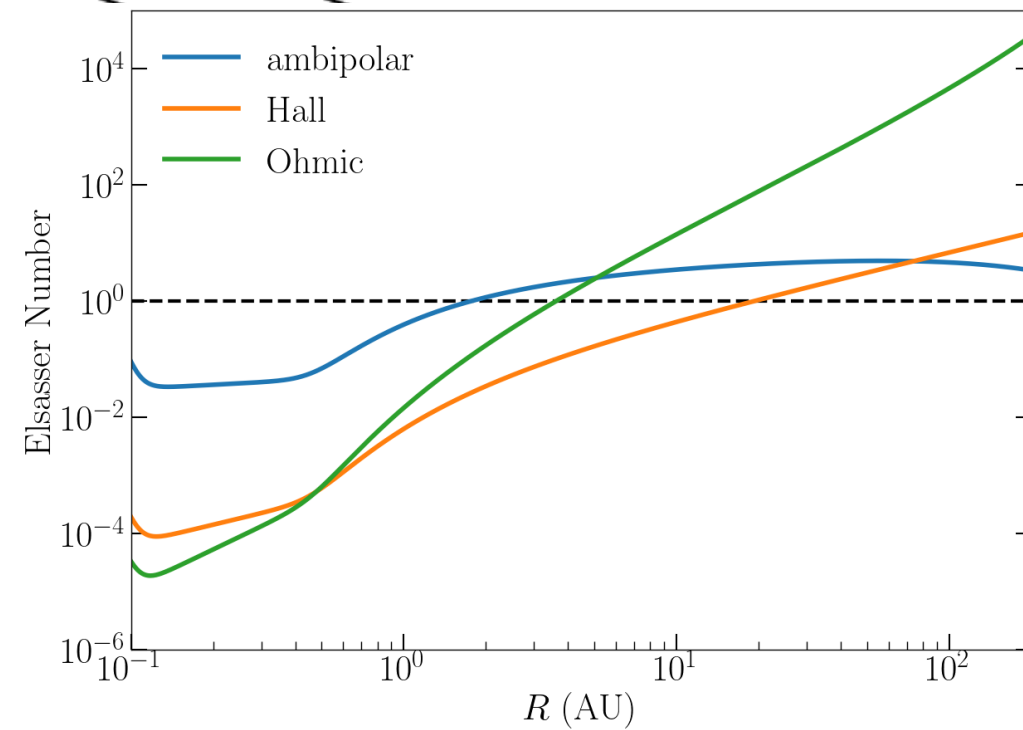
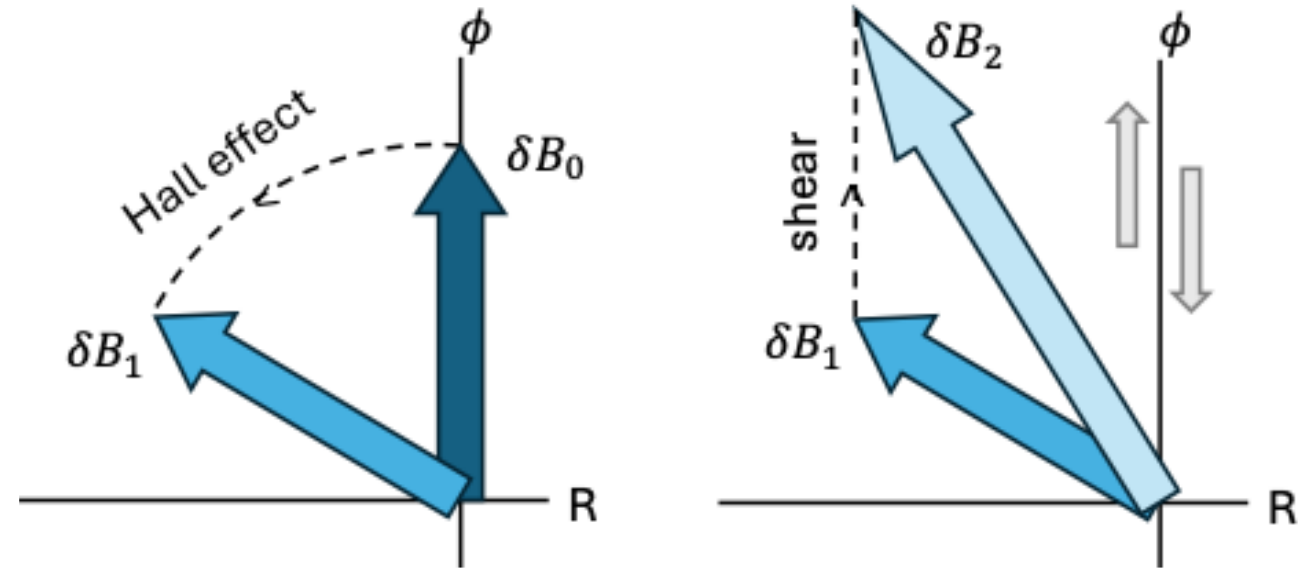




MRI dead zone

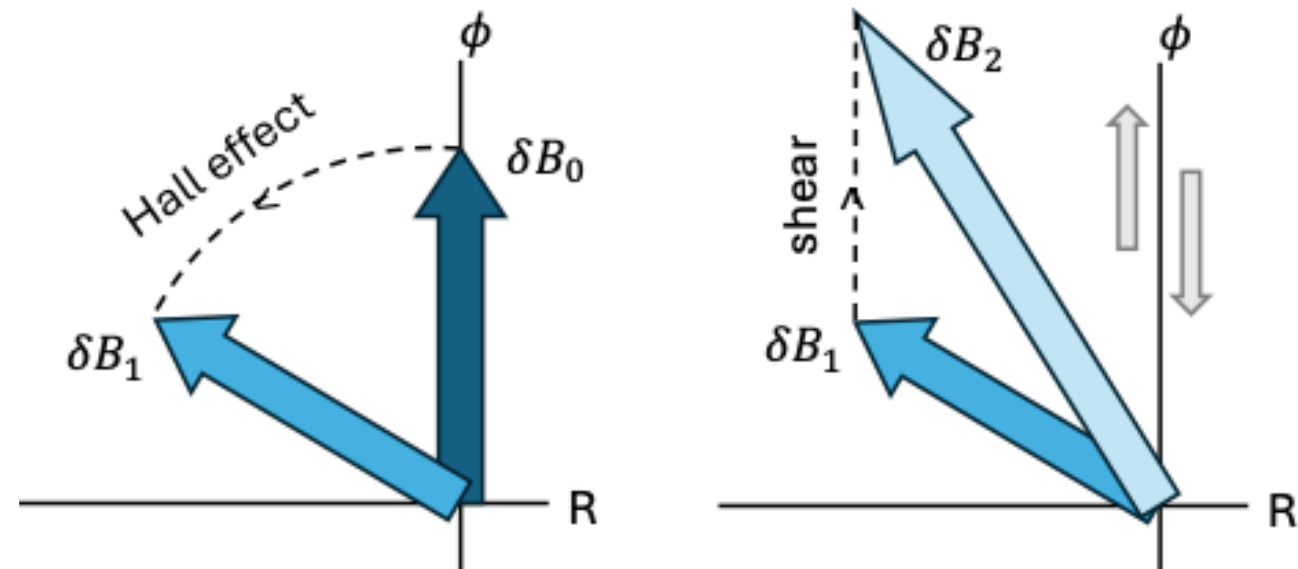




Hall Shear Instability

1. Hall effect rotates perturbations into the shear flow ($B_\phi \rightarrow B_R$)
2. Shear stretches perturbations ($B_R \rightarrow B_\phi$)

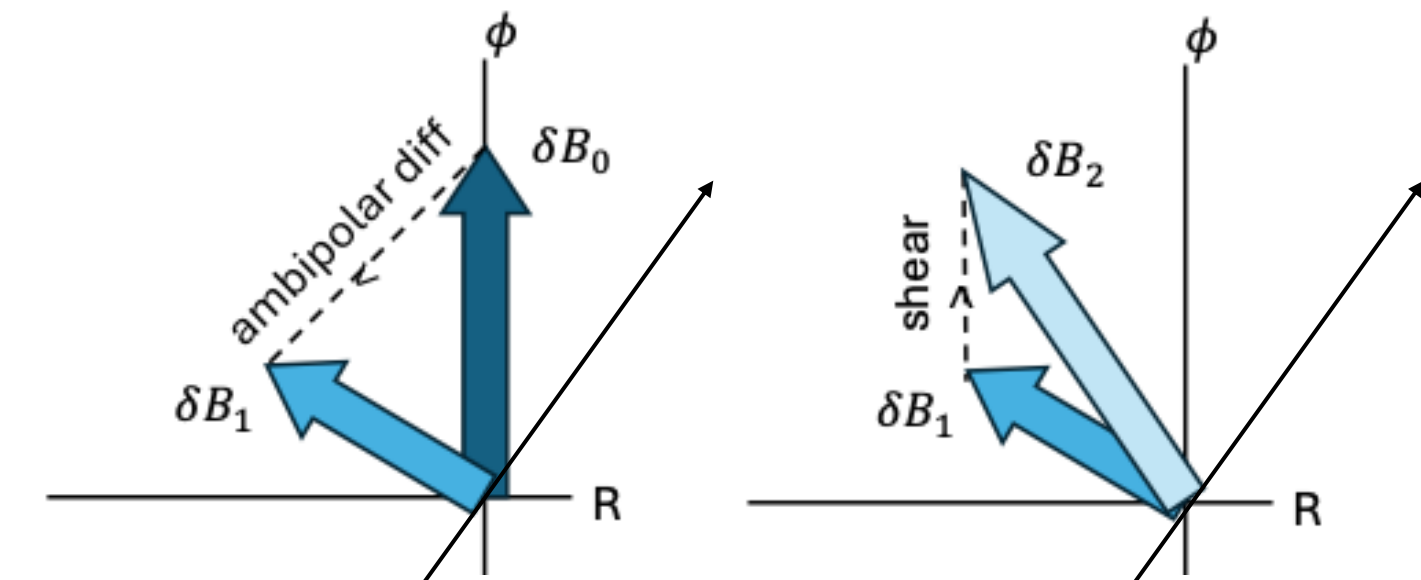
Vertical field orientation determines stability



Hall Shear Instability

1. Hall effect rotates perturbations into the shear flow ($B_\phi \rightarrow B_R$)
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Vertical field orientation determines stability

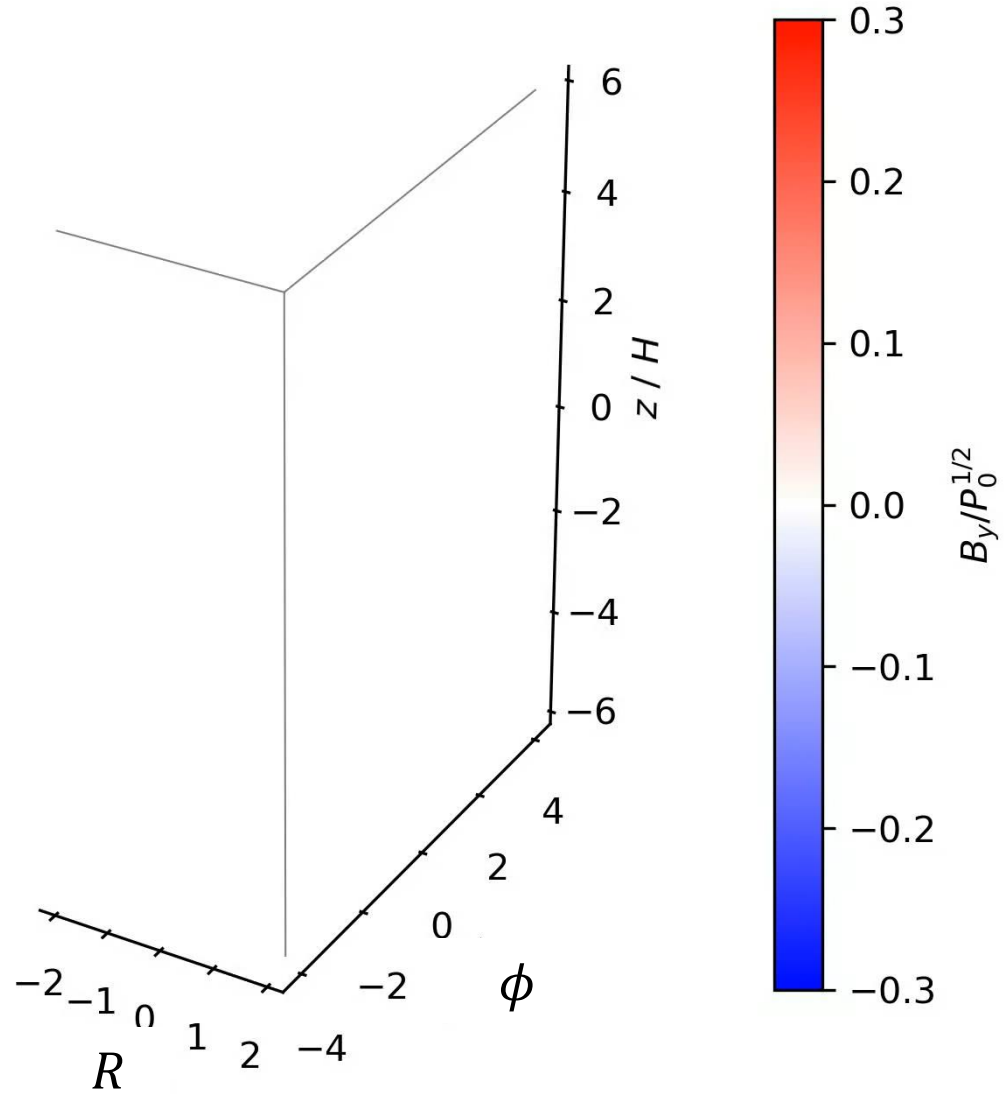


Ambipolar Diffusion Shear Instability

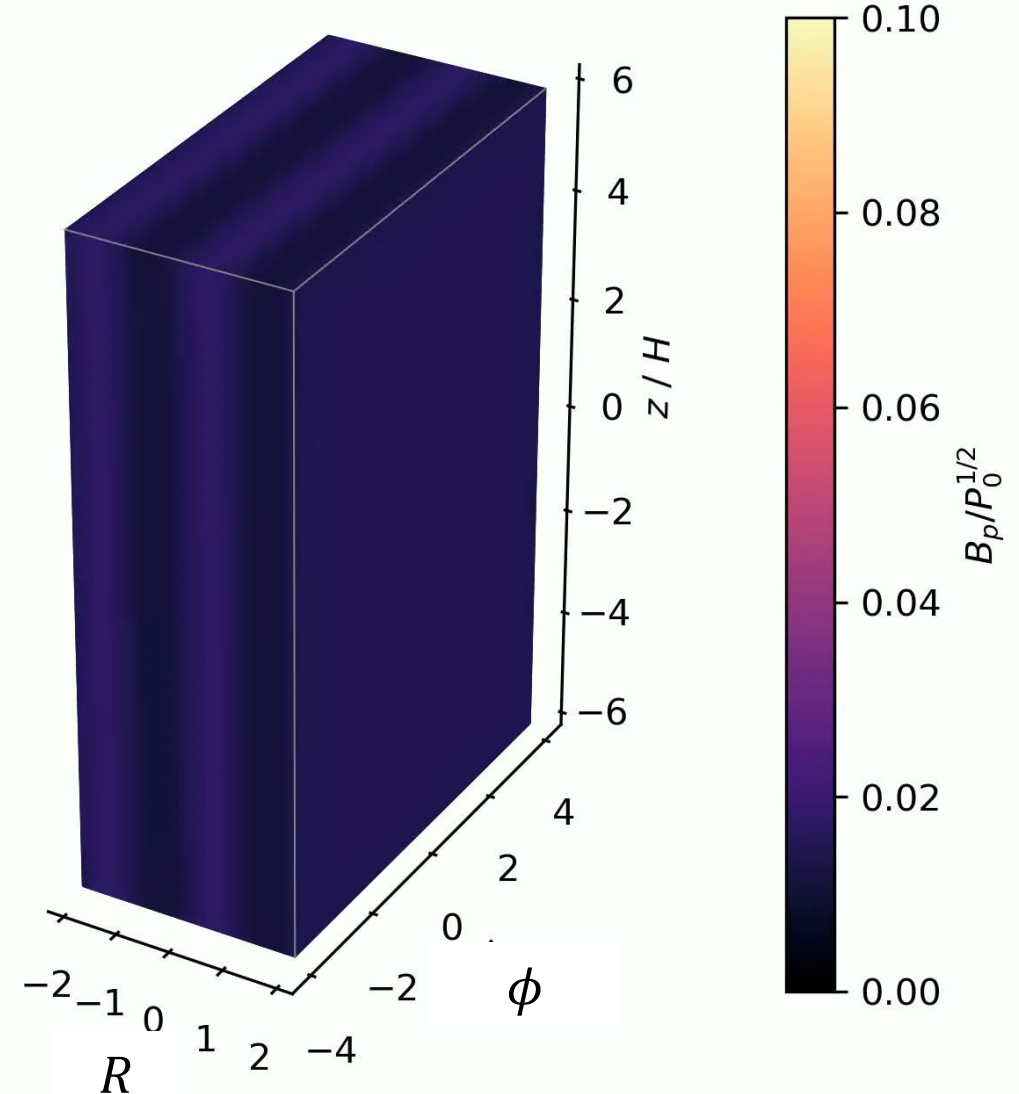
1. AD retains perturbations perpendicular to the background magnetic field ($B_\phi \rightarrow B_R$)
2. Shear stretches perturbations ($B_R \rightarrow B_\phi$)

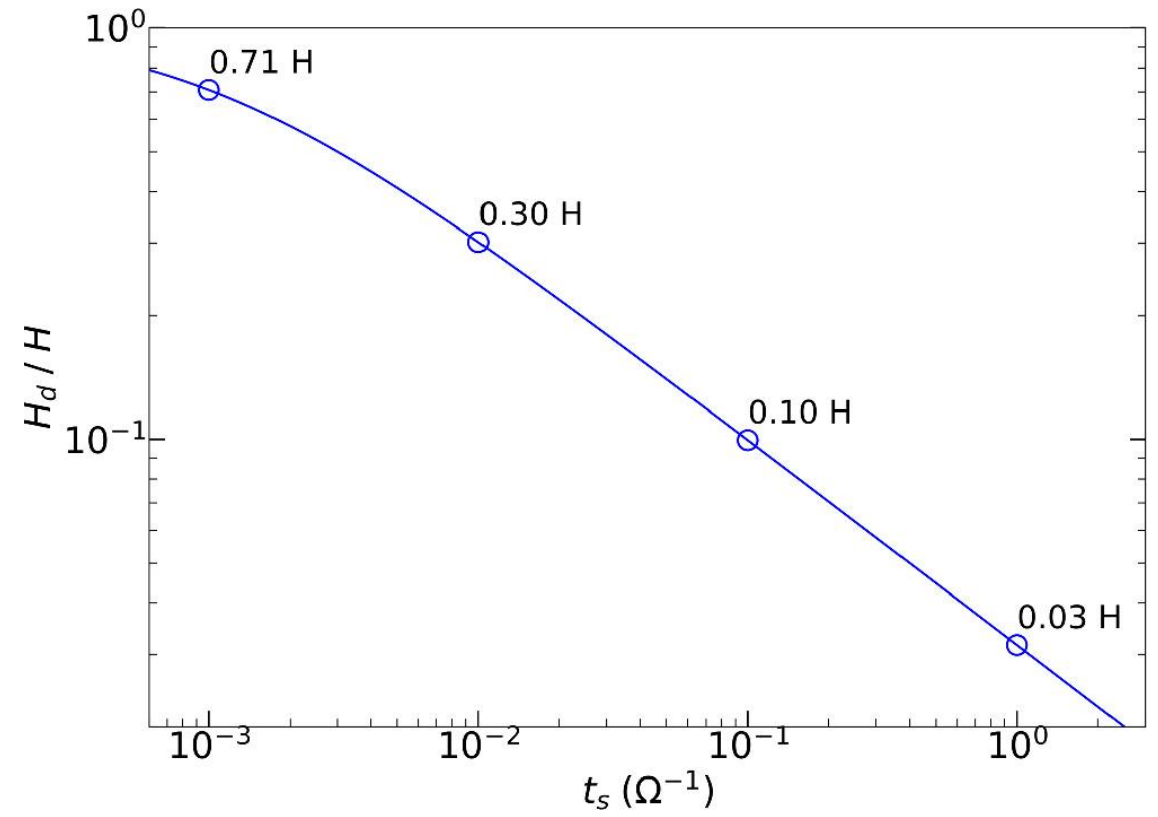
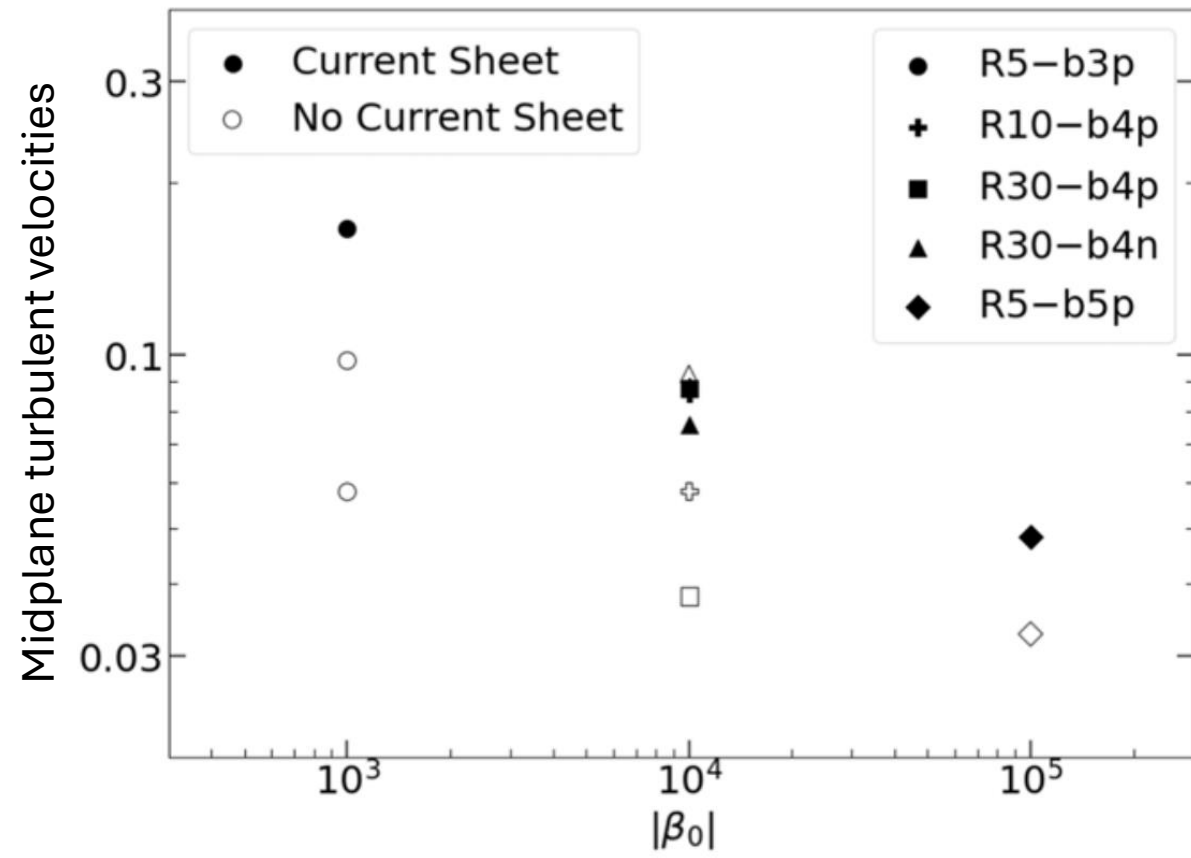
Azimuthal field determines stability

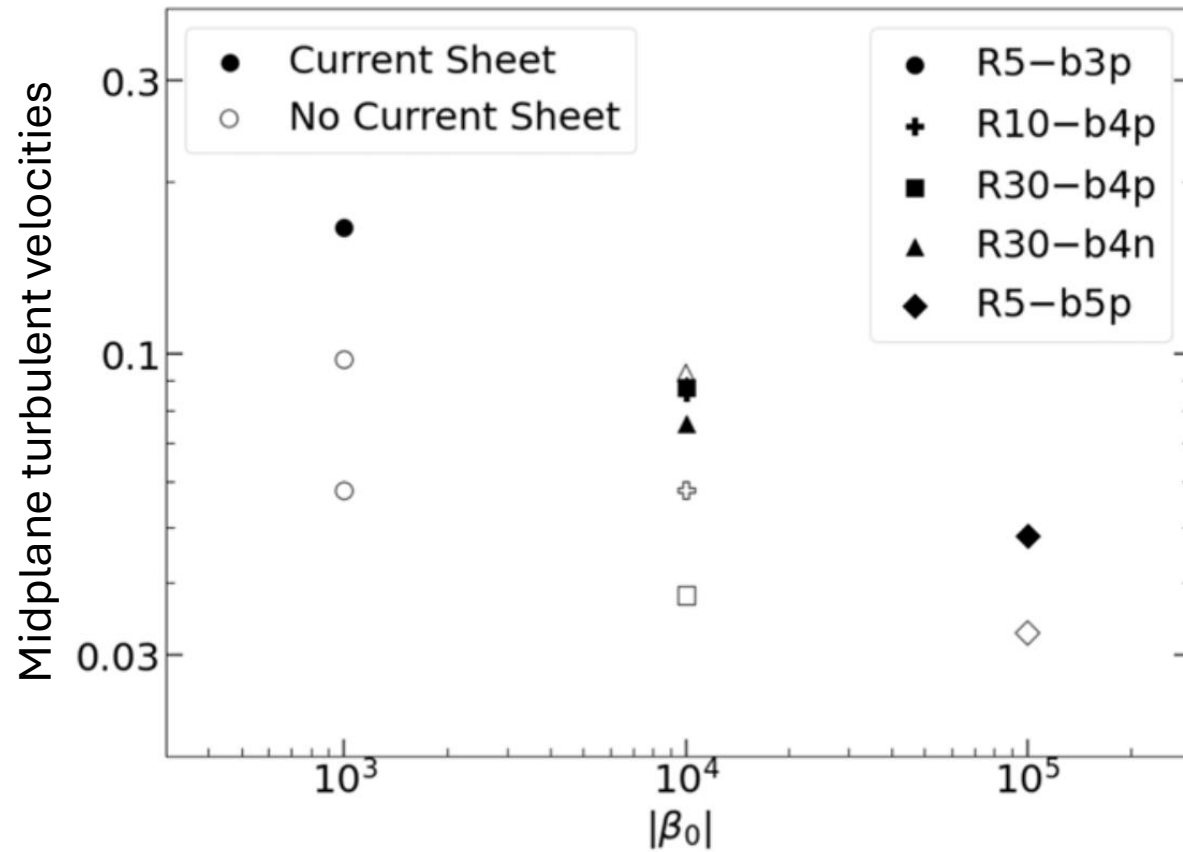
Azimuthal Field



Radial + Vertical Field

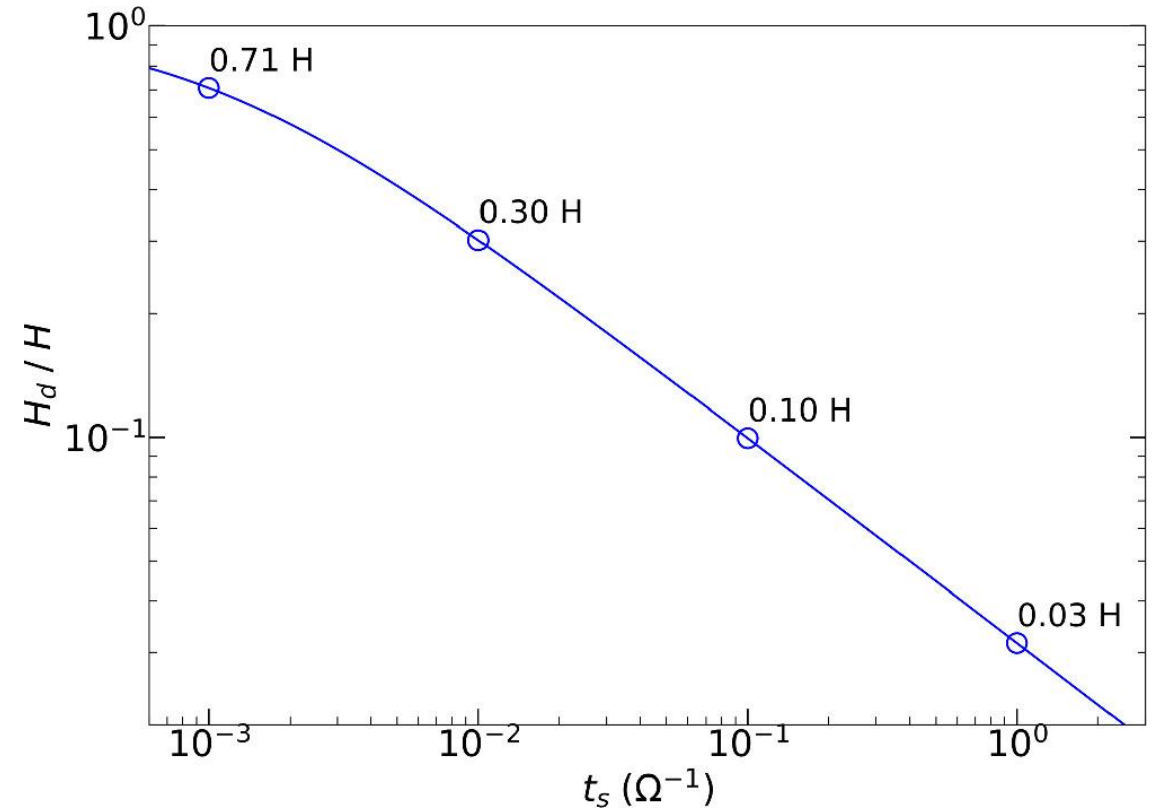






Midplane turbulent velocities:
 $\delta v \sim 10^{-2} c_s$, or **16 – 75 m/s**

Turbulent accretion:
 $\alpha \sim 10^{-3}$, or $\dot{M} \sim 5 \times 10^{-9} M_{\odot}/yr$

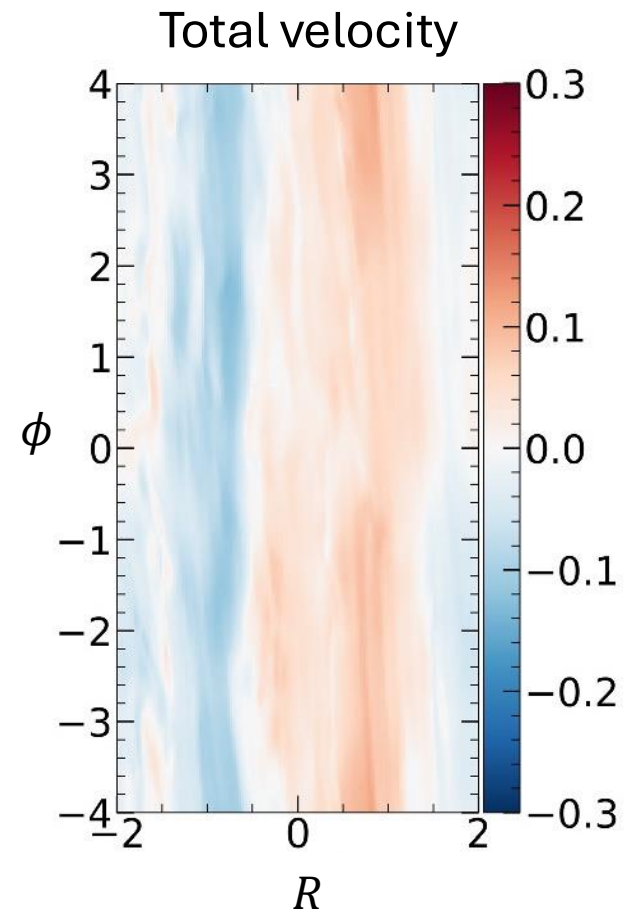
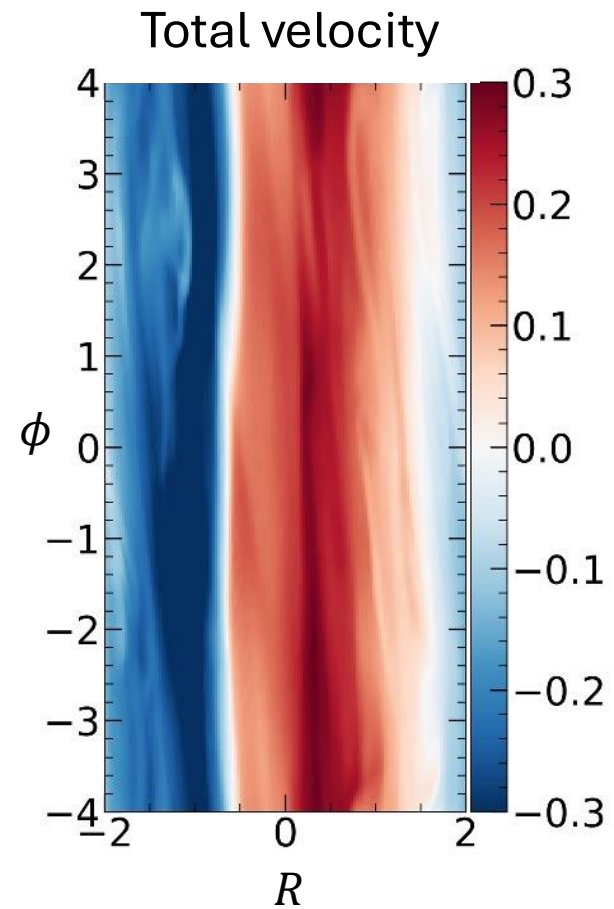


Small dust grains are vertically
 well-mixed with the gas

$v_{turb} > v_{frag} \sim 25 - 50 \text{ m/s}^*$

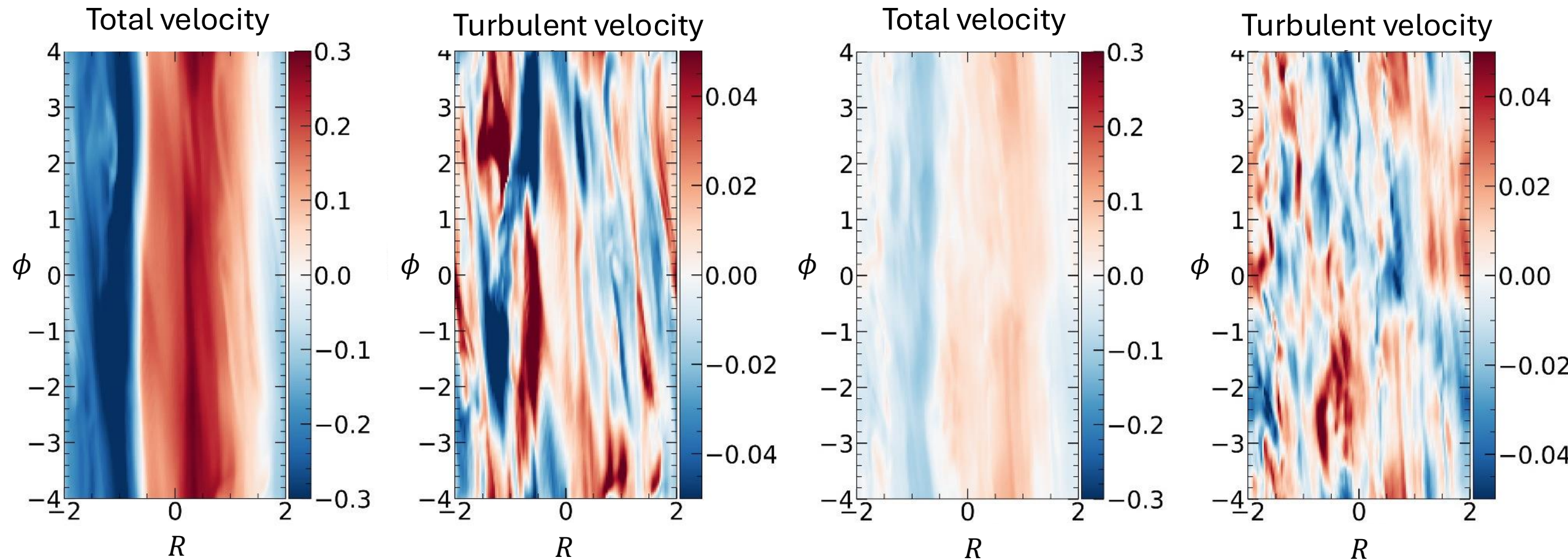
Hall Shear Instability

no HSI present

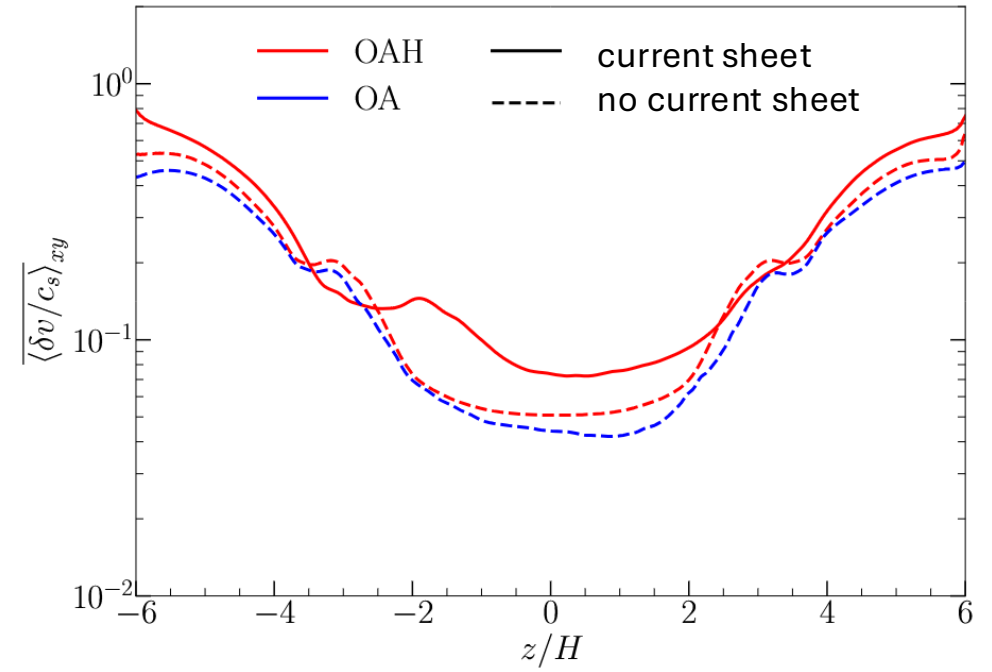


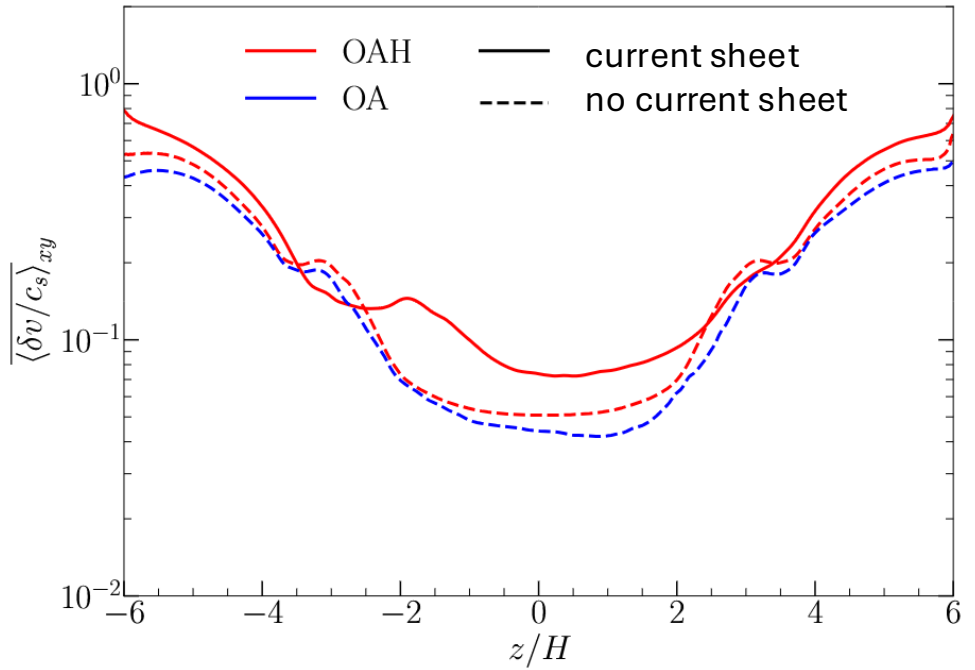
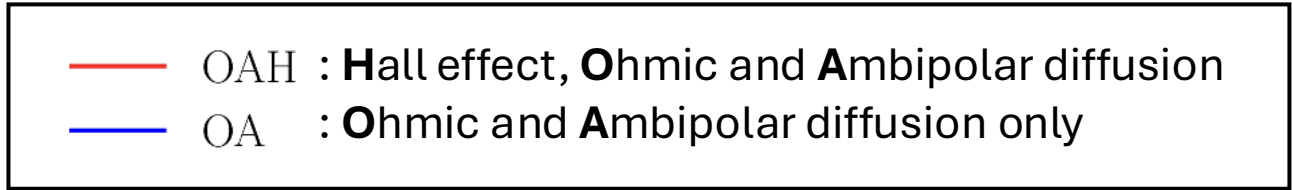
Hall Shear Instability

no HSI present

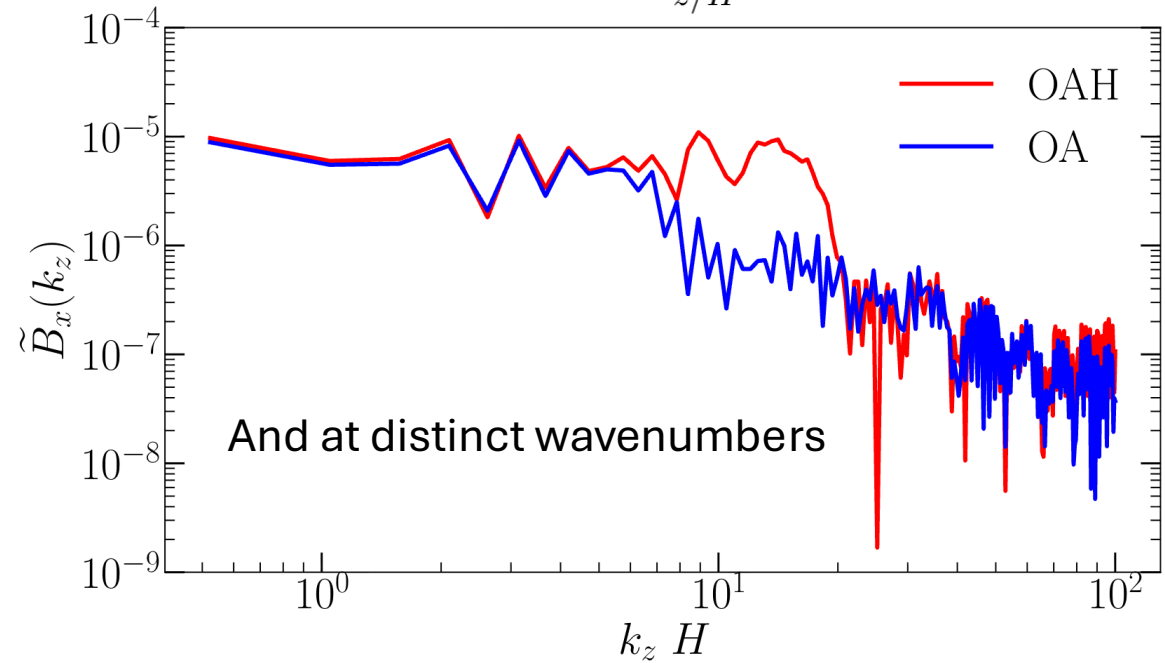
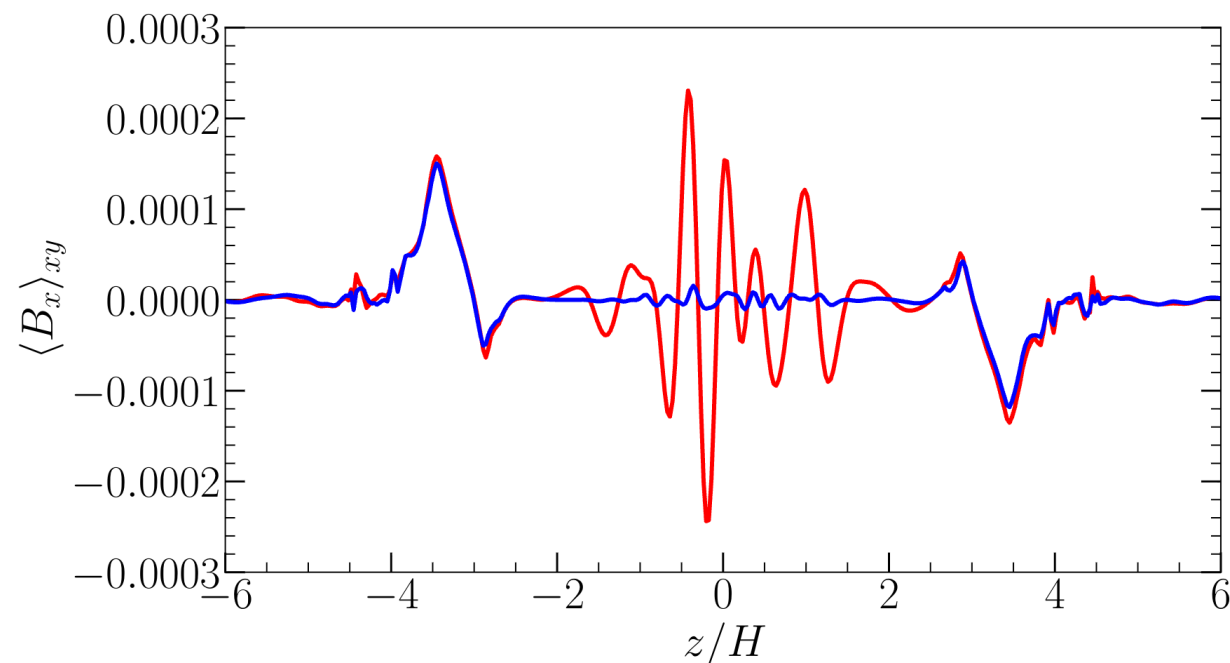


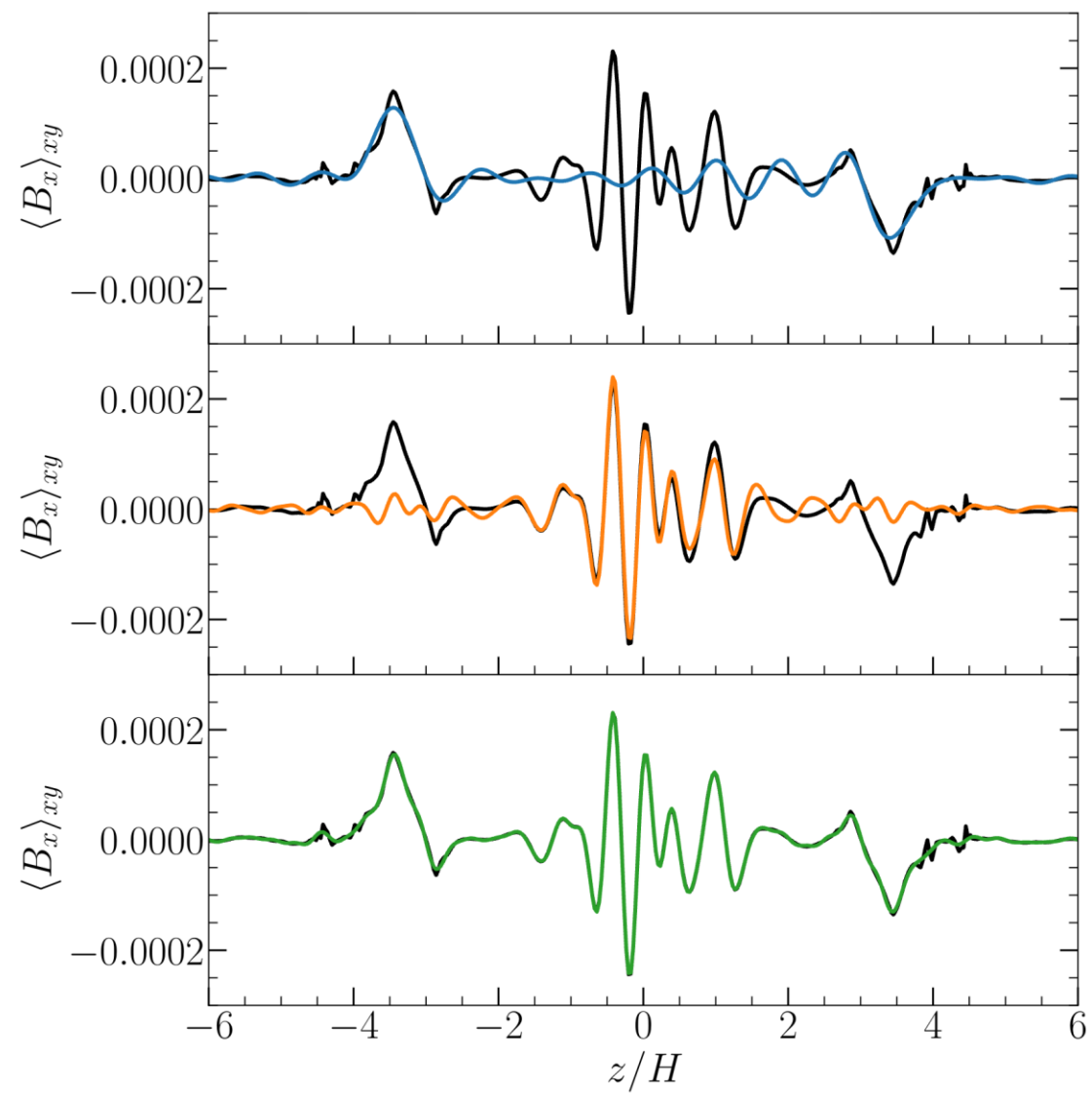
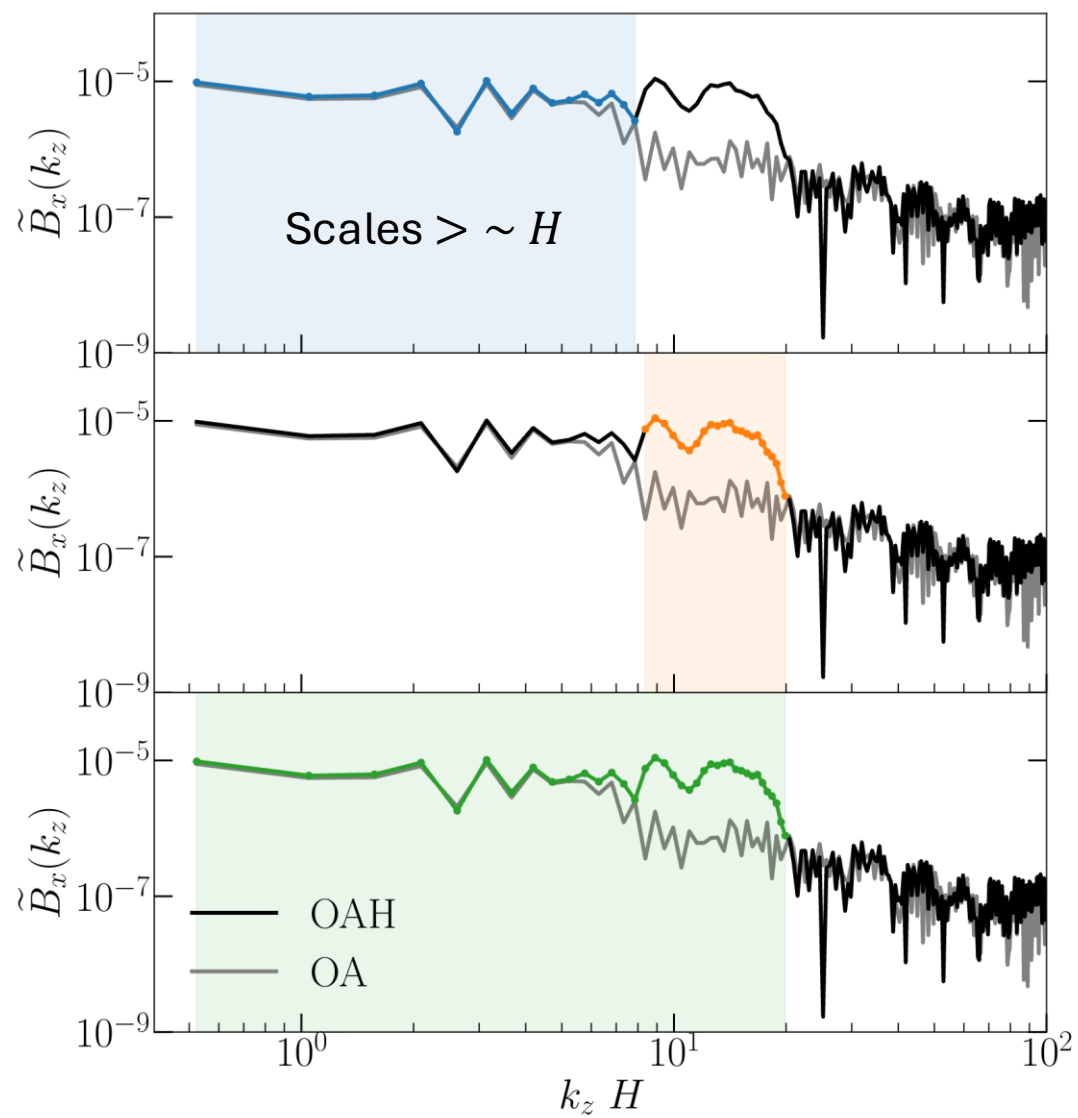
- OAH : **H**all effect, **O**hmic and **A**mbipolar diffusion
- OA : **O**hmic and **A**mbipolar diffusion only

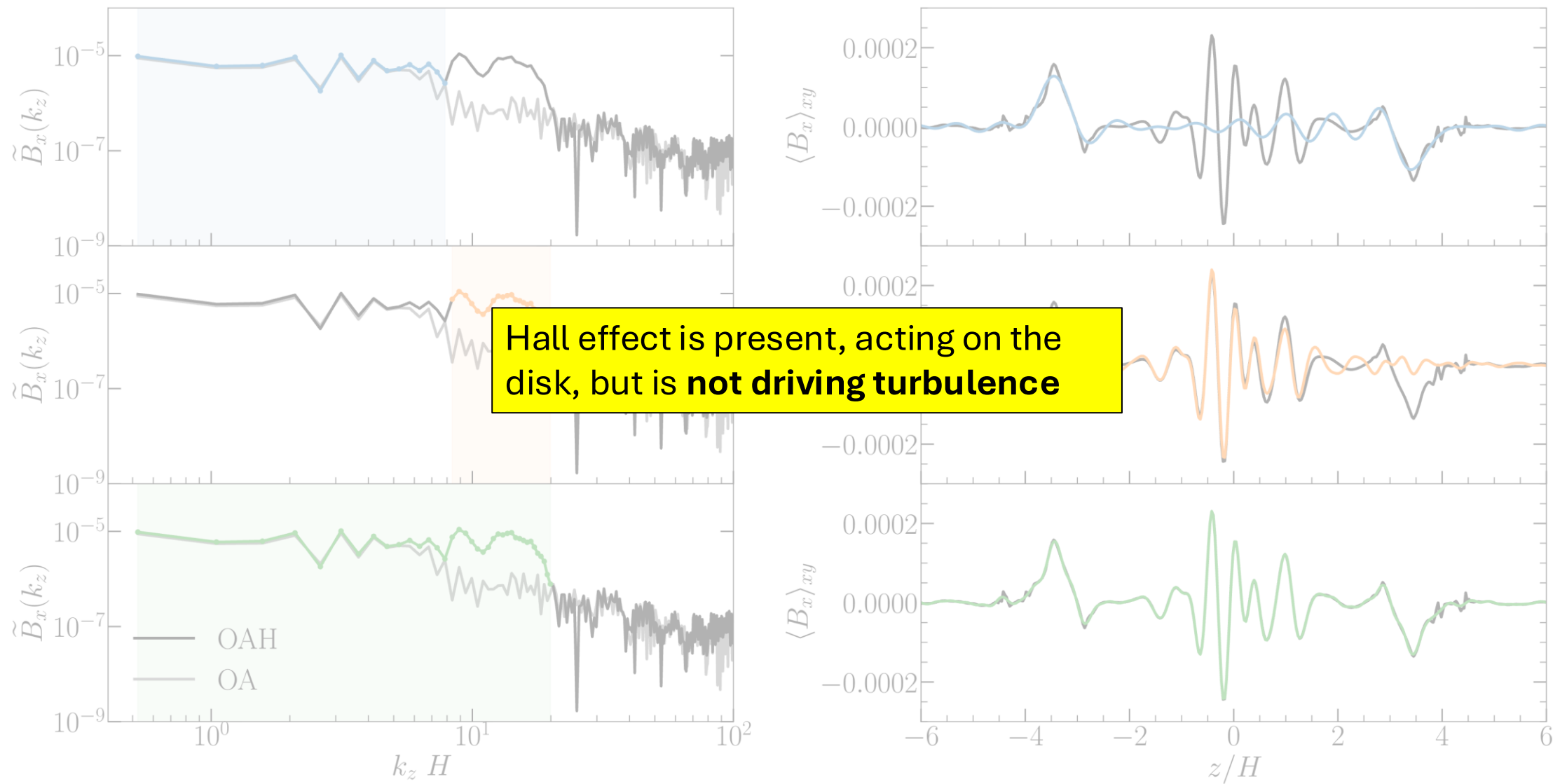




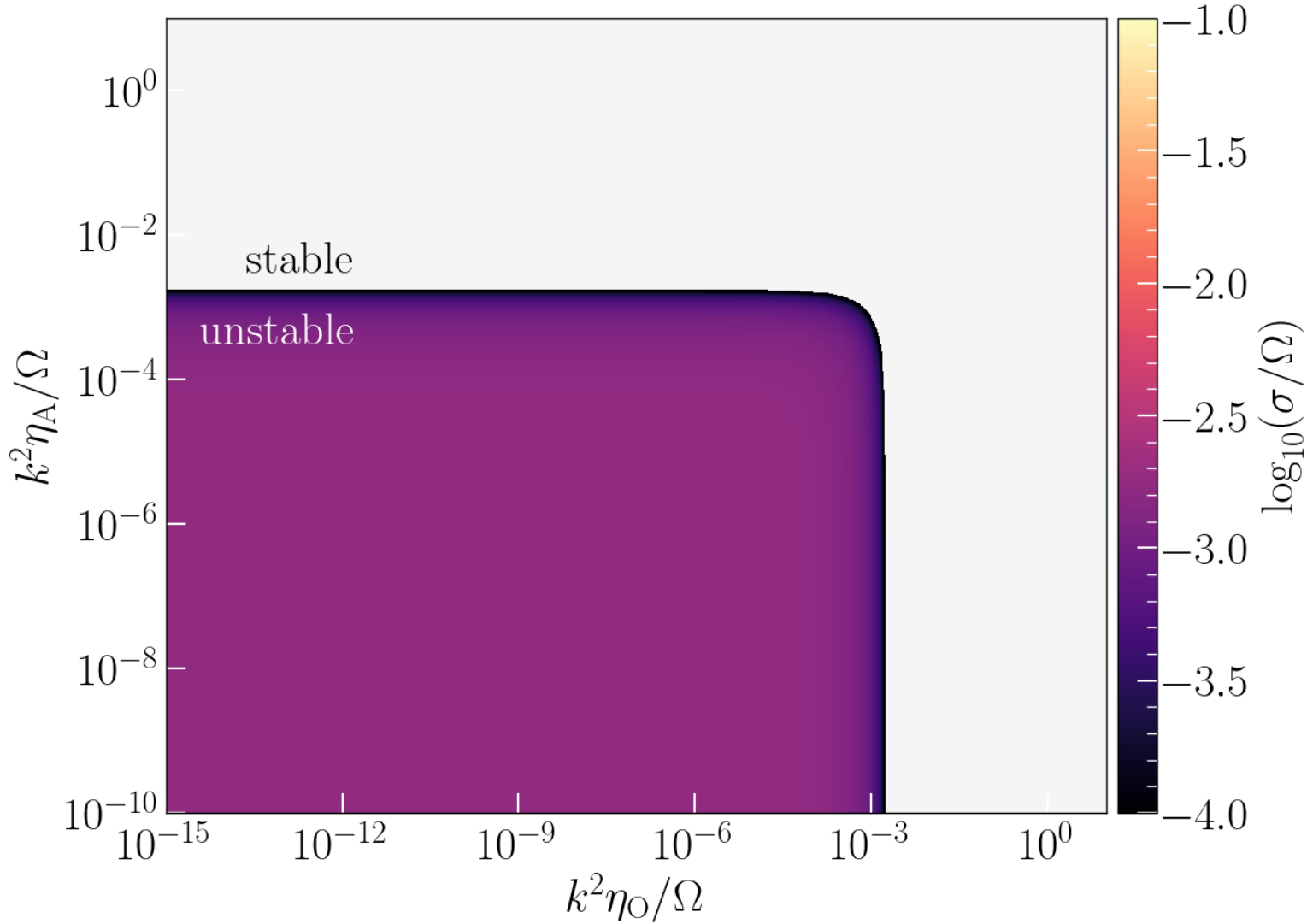
Hall effect drives magnetic field growth at the midplane







Linear Theory Predictions



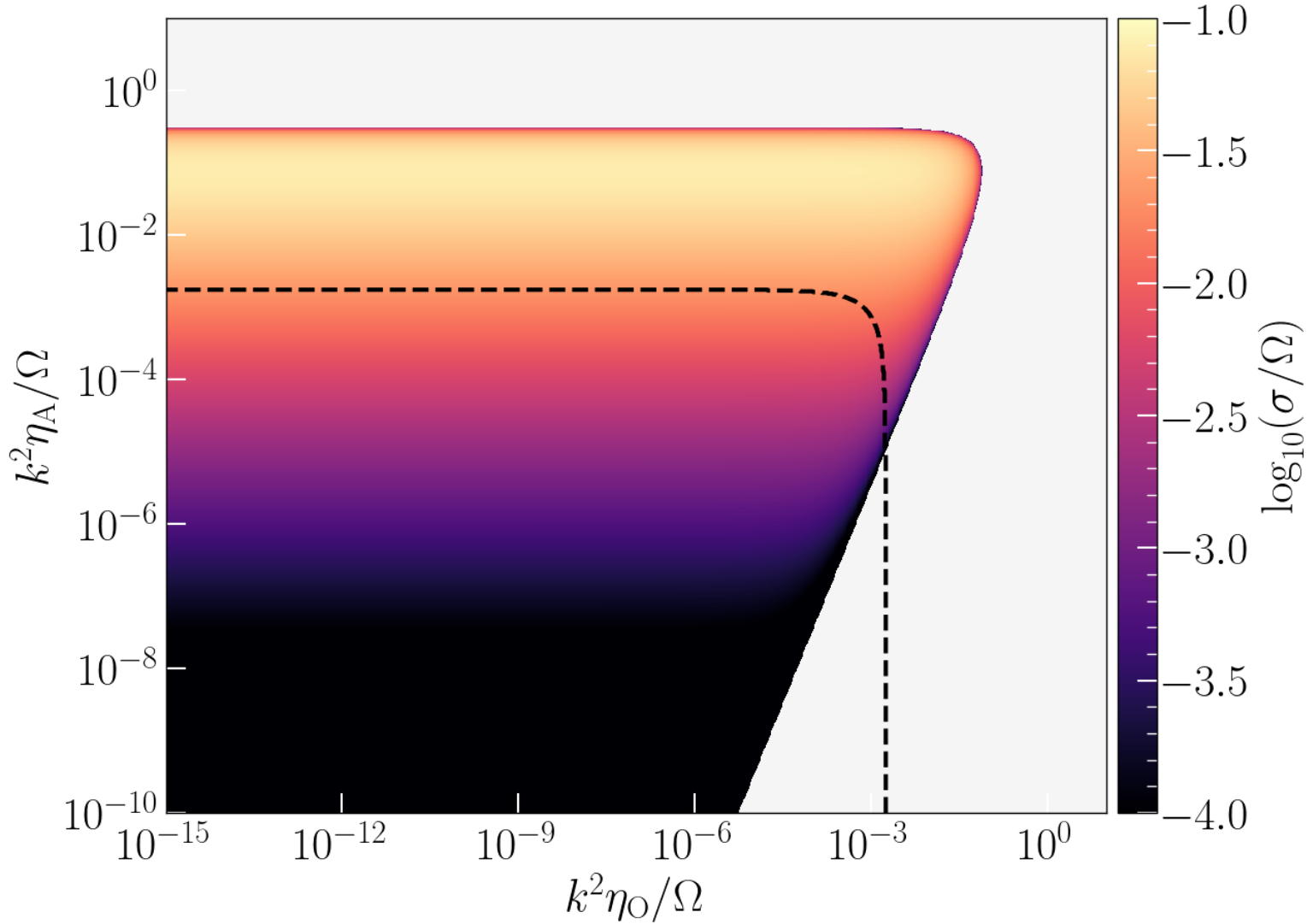
MRI + diffusion

- No AD-shear coupling
- Ambipolar diffusion only stabilizes the disk

Outcome

- Growth rates are damped
- MRI is stabilized by strong ohmic diffusion
- MRI is stabilized by strong ambipolar diffusion

Linear Theory Predictions



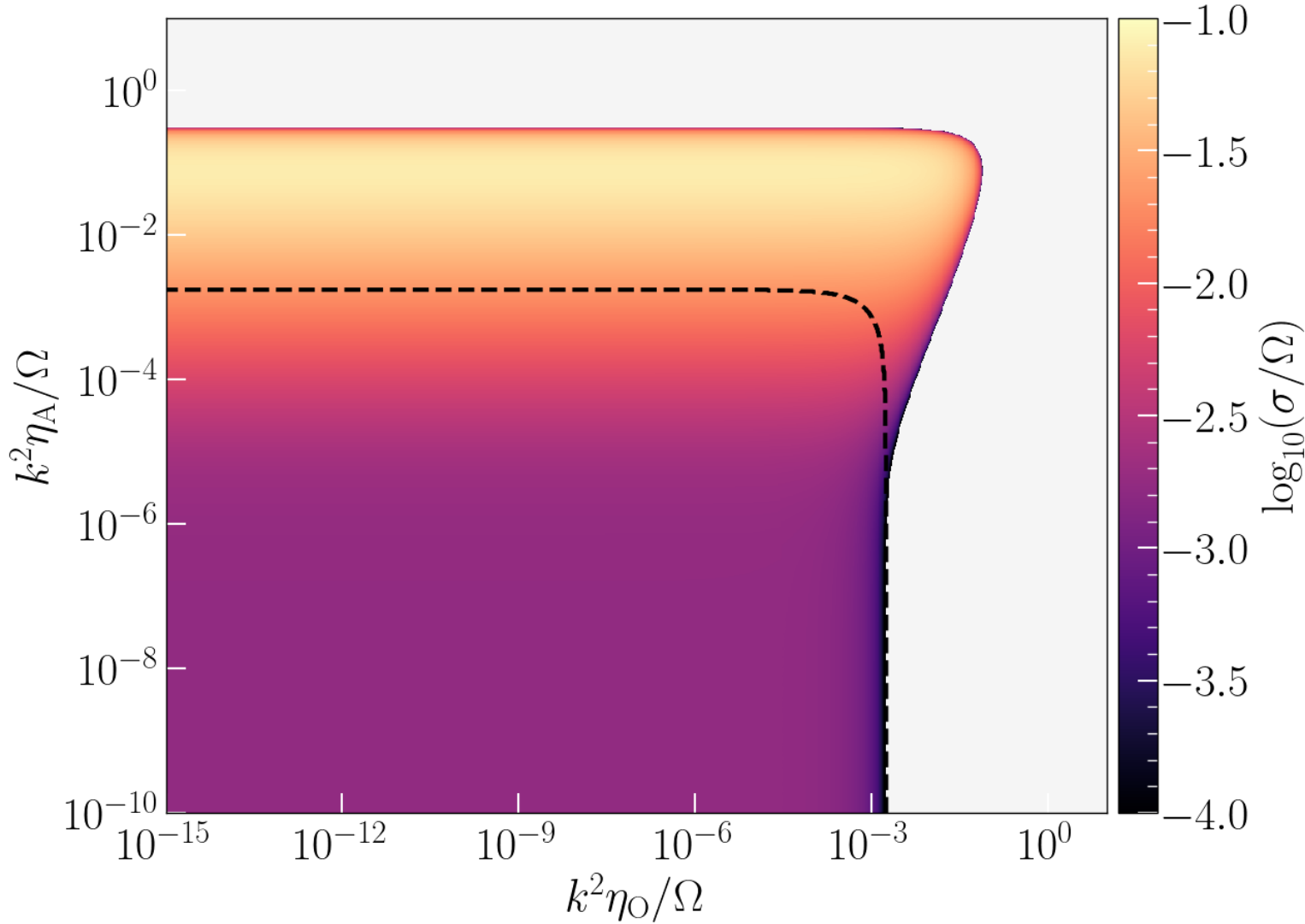
ADSI + diffusion

- $kv_A \ll \Omega$
- Magnetic tension force always damped $\rightarrow$ no MRI

Outcome

- Strong ambipolar diffusion enhances the instability
- Still damped by strong enough diffusion

Linear Theory Predictions



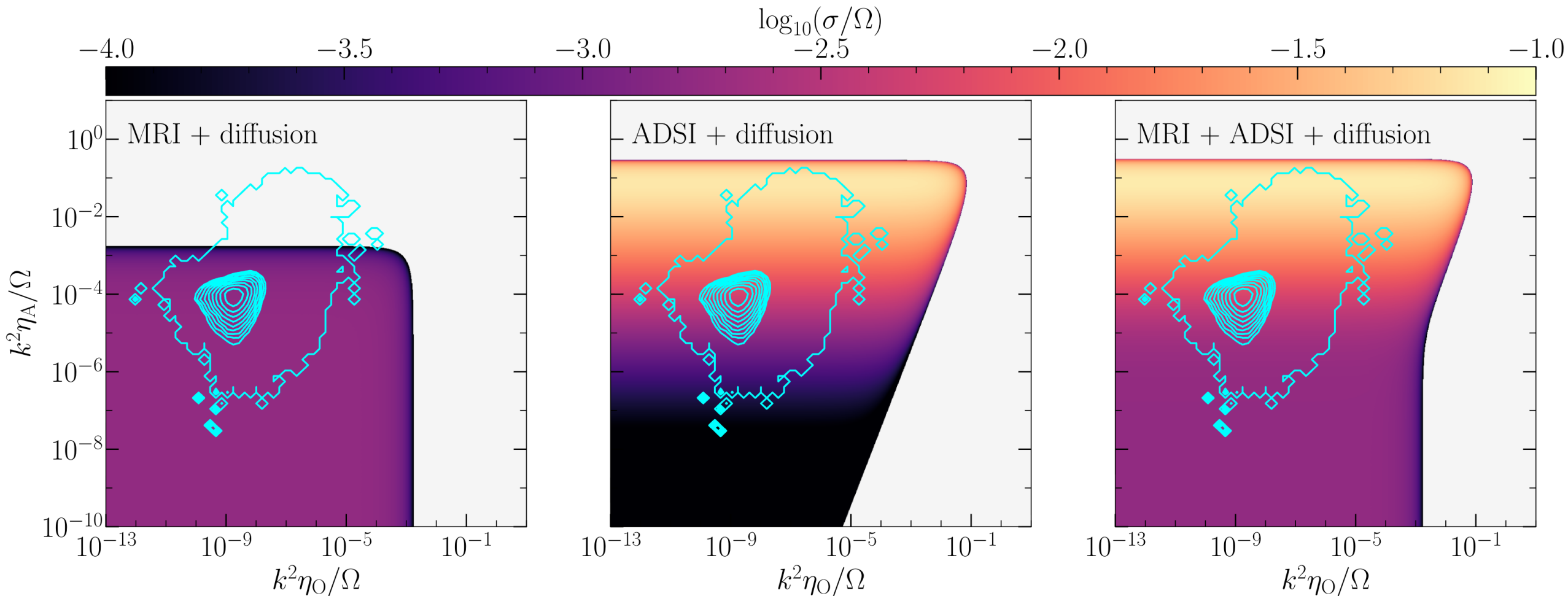
MRI + ADSI + diffusion

- Full dispersion relation (no Hall effect)

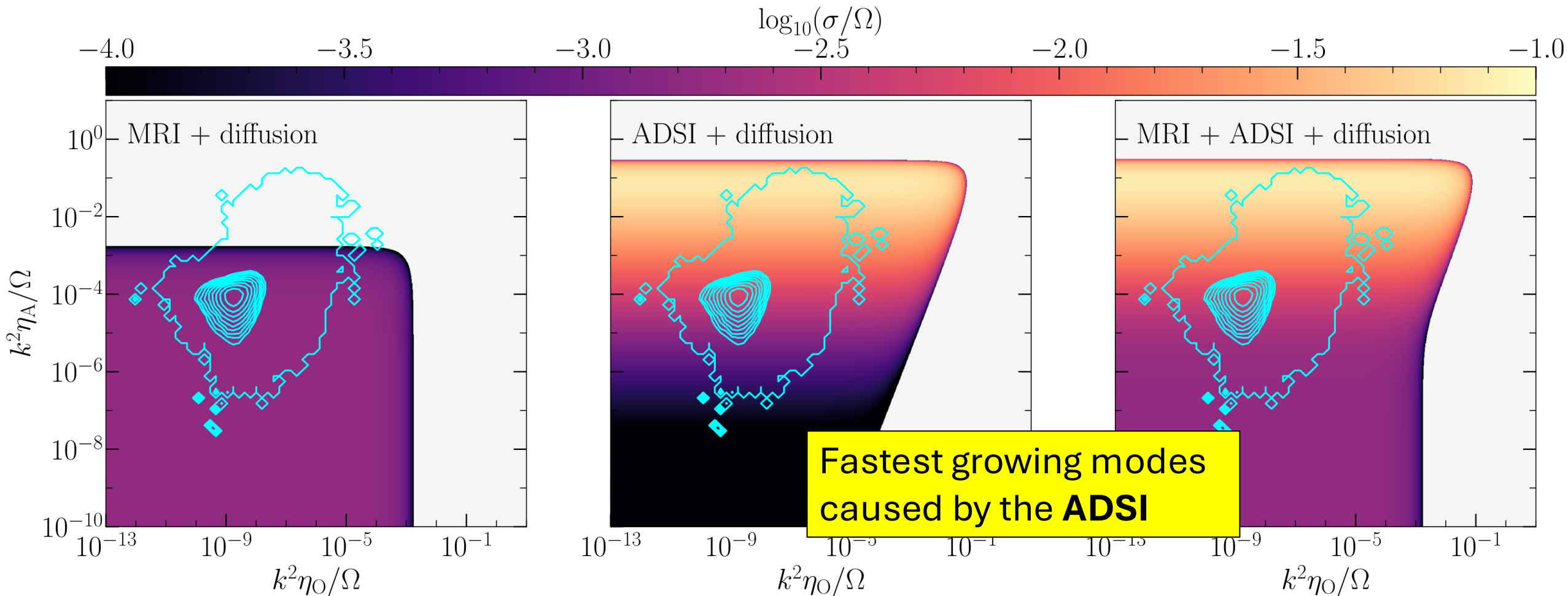
Outcome

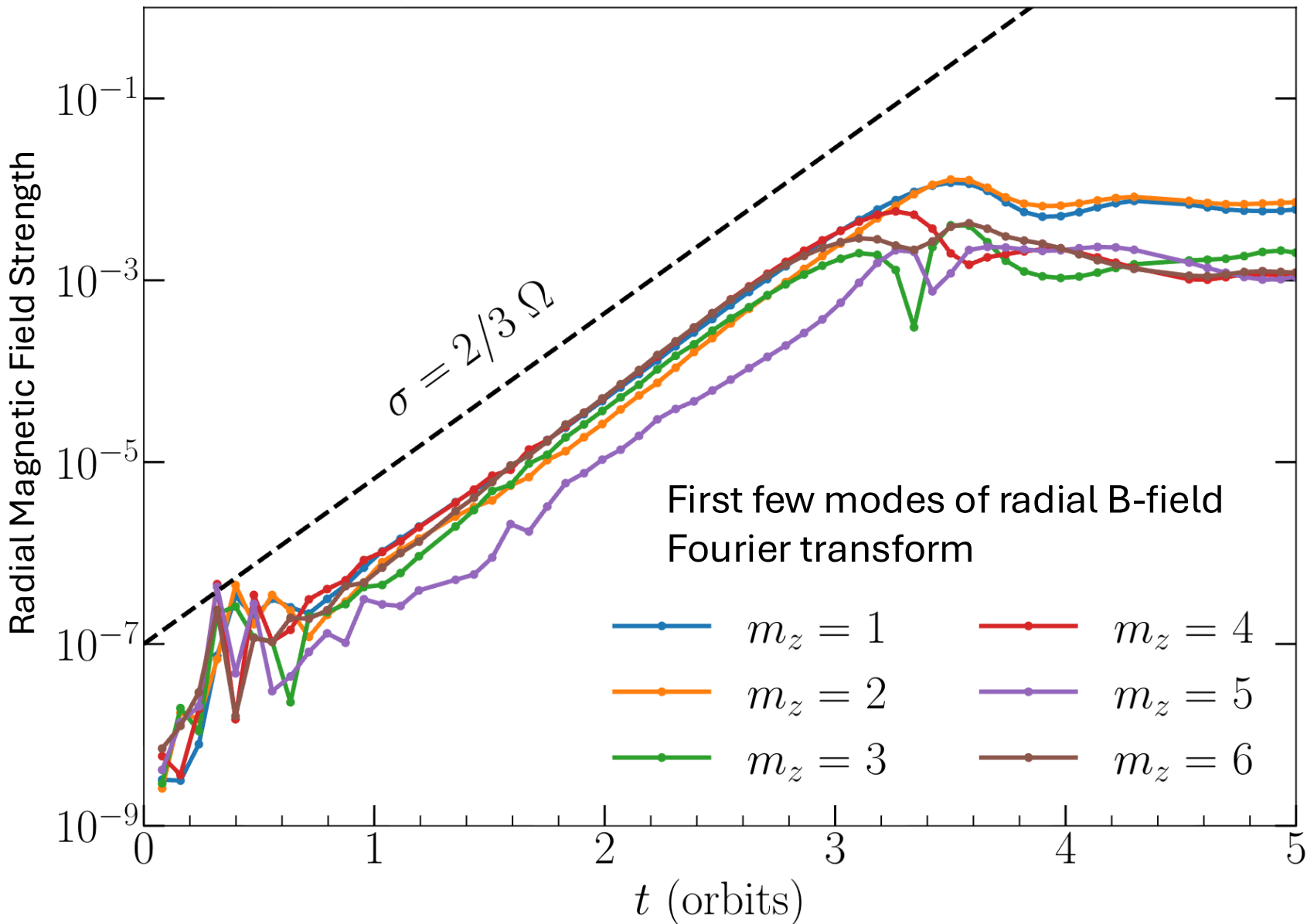
- Strong ambipolar diffusion enhances the combined instability
- Still damped by strong enough diffusion

What are the simulation expected growth rates?



What are the simulation expected growth rates?

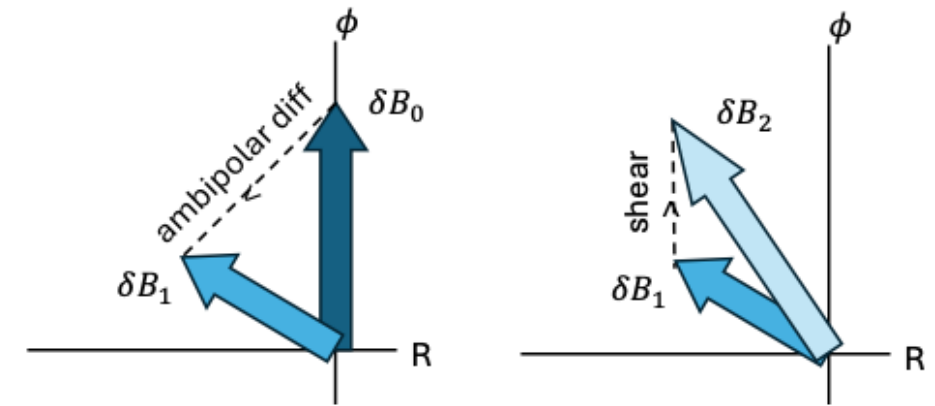




Instability	Growth rate
Actual Simulation	0.66 Ω
MRI + ADSI	0.60 Ω
HSI	0.36 Ω

3D fully non-ideal MHD SB simulations

- > Turbulent velocity fluctuations of 16-75 m/s at the disk midplane
- > May cause collisional dust destruction and disrupt Streaming Instability clumping



Appears Difficult to Suppress MHD-Instabilities

- > Ambipolar diffusion (ADSI) can kickstart growth of magnetic fluctuations
- > Hall effect appears to have limited role in driving turbulence

